UNIT–2 AGILE MINI-PROJECT REPORT

STUDENT RECORD AND ACADEMIC MANAGEMENT PLATFORM

Submitted as a Mini-Project Report

Subject: Agile Project Planning

Name: Kartik Sable

PRN: 2126UMLM2540

Branch: Artificial Intelligence and Machine Learning

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1. Introduction

Agile is an iterative and flexible approach to software development. It divides a large project into smaller parts and delivers working features in short development cycles called Sprints. This report presents the Unit-2 Agile Mini-Project planning for a Student Record and Academic Management Platform.

The proposed platform is intended to help educational institutions manage student information, courses, attendance, academic performance, authentication, notifications, and reports through a web-based application. The report follows the activities and deliverables described in the supplied Unit-2 document.

2. Study of the Business Problem and Product Vision

Educational institutions maintain a large amount of student and academic information. When records are handled through disconnected or manual processes, searching, updating, and preparing reports can take time. A centralized platform can make these activities easier to manage.

2.1 Product Vision, Mission and Scope

Product Vision: Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.

Mission: Provide a centralized system that supports accurate academic record management and makes important student information easier for authorized users to access and maintain.

2.2 Stakeholders

- Students – use the platform to view relevant profile, course, attendance, and result information, subject to access permissions.
- Faculty Members – manage attendance and marks and access course-related records.
- Administrative Staff – maintain student and course records and prepare institutional reports.
- Institution Management – review academic information and reports for administration and planning.
- System Administrator – manage user access, security, and system configuration.

2.3 Business and System Requirements

• Functional requirements: student registration and profile management; course creation and assignment; attendance recording and reports; marks entry, grade calculation and result sheets; login, logout and password reset; role-based access; notifications; and reporting.

• Non-functional requirements: security, usability, reliability, performance, maintainability, and scalability. These qualities should guide design and testing.

2.4 Minimum Viable Product (MVP) and Roadmap

A practical MVP can begin with user authentication, student registration and search, and basic course and attendance management. Later releases can add marks and grades, notifications, advanced reports, and academic analytics. This is a proposed sequencing of the capabilities listed in the Unit-2 brief.

3. Agile Product Vision and Success Criteria

The Agile product vision converts the broad purpose of the platform into an outcome that can guide the Scrum Team. The main objective is to deliver useful academic-management features incrementally while keeping stakeholder needs and product quality in focus.

• Primary users: students, faculty members, administrative staff, management, and system administrators.

• Business benefits: centralized academic records, reduced repetitive manual work, easier record retrieval, more consistent attendance and marks management, and timely reporting.

• Expected capabilities: secure access, student and course management, attendance, marks and grades, notifications, and reports.

• Project constraints: limited student-team time, available development resources, changing stakeholder feedback, and the need to protect academic information.

Suggested measurable success criteria: authorized users can complete their assigned tasks; student and course records can be created, searched, updated, and retrieved; attendance and marks are stored correctly; invalid login attempts are handled safely; and required reports can be generated. Exact numerical targets should be agreed with the project guide or stakeholders.

4. Scrum Team Structure and Responsibilities

The Unit-2 document identifies three Scrum role groups: Product Owner, Scrum Master, and Development Team. The team should assign these responsibilities clearly. One student may take more than one responsibility in a small academic project, while the duties should remain understood.

Role	Main responsibilities
Product Owner	Defines requirements; prioritizes the Product Backlog; represents stakeholder interests; clarifies user stories; accepts completed work.
Scrum Master	Facilitates Scrum ceremonies; helps remove impediments; supports collaboration; encourages the team to follow Agile practices.
Development Team	Designs the application; writes code; tests features; prepares documentation; demonstrates completed work.

5. Project Epics

An Epic is a high-level functional area that can be divided into smaller user stories. The Unit-2 brief identifies seven epics for the platform.

Epic	Purpose and included features
1. Student Management	Student registration, profile management, student search, student update, and student deletion.
2. Course Management	Add courses, update courses, assign students, and view course details.
3. Attendance Management	Record attendance, view attendance, and generate attendance reports.
4. Marks and Academic Performance	Enter marks, update marks, calculate grades, and generate result sheets.
5. Authentication and Security	User login, logout, password reset, and role-based access control.
6. Notifications	Attendance alerts, examination notifications, and result notifications.

Epic	Purpose and included features
7. Reporting	Student reports, attendance reports, performance reports, and course reports.

Each Epic should be refined into user stories before implementation. The team can add technical tasks and testing tasks to the Sprint Backlog when needed.

6. User Stories and Acceptance Criteria

A user story describes a requirement from the viewpoint of a user. The standard format used here is: “As a [user], I want [capability], so that [benefit].” Acceptance criteria describe conditions that must be satisfied before a story can be considered complete.

ID / Epic	User story	Acceptance criteria (summary)
US-01 Authentication	As a registered user, I want to log in so that I can access permitted features.	Login page loads; valid credentials permit access; invalid credentials show an error; successful login opens the appropriate dashboard; session handling works.
US-02 Student Management	As an administrator, I want to add a student so that the student record is stored centrally.	Required fields are validated; a valid record is saved; confirmation is shown; duplicate or invalid data is handled.
US-03 Student Management	As an authorized user, I want to search for a student so that I can find a record quickly.	Search accepts an appropriate identifier or name; matching records are displayed; no-match results are communicated.
US-04 Student Management	As an authorized user, I want to update student details so that records remain current.	Existing details are displayed; permitted fields can be edited; valid changes are saved; confirmation is shown.
US-05 Course Management	As an administrator, I want to add a course so that it can be assigned to students.	Course details are validated; valid course is saved; course details can be viewed.
US-06 Attendance Management	As a faculty member, I want to record student attendance so that attendance records remain accurate and updated.	Faculty can select a course and lecture date; attendance can be marked; records are saved; attendance reports can be generated.
US-07 Marks and Performance	As a faculty member, I want to enter student marks so that academic performance can be maintained.	A valid course/student context is selected; marks are checked against allowed limits; saved marks can be viewed or updated.
US-08 Notifications	As a student, I want to receive attendance and examination alerts so that I can stay informed.	Relevant notification is generated; intended user can view it; notification content is clear.
US-09 Reporting	As an administrator, I want to generate academic reports so that I can review institutional records.	User selects report type and relevant filters; report data is displayed; report reflects stored records.

The detailed criteria above are report-ready examples derived from the features and sample criteria in the supplied Unit-2 brief. The project team should confirm field-level validation rules with its guide.

7. Product Backlog and Prioritization

The Product Backlog is the ordered list of work required for the product. The Unit-2 brief recommends prioritizing items according to business value, user importance, technical dependencies, and implementation complexity.

Priority	Epic / item	Reason for priority
High	Authentication – User Login	Provides controlled access and is a foundation for other modules.
High	Student Management – Add Student	Core record-management capability.
High	Student Management – Search Student	Makes stored records accessible to authorized users.
High	Attendance – Record Attendance	Core academic activity identified in the Unit-2 example.
Medium	Course Management – Add Course	Supports course organization and assignment.
Medium	Reporting – Attendance Report	Provides useful visibility into attendance records.
Medium	Notifications – Attendance Alert	Helps communicate attendance-related information.
Low	Dashboard – Academic Analytics	Can be enhanced after core data and modules are available.

Backlog order should be reviewed whenever stakeholder feedback, dependencies, or project constraints change.

8. Story Point Estimation

Story Points are a relative estimate of the effort, complexity, and uncertainty involved in completing a user story. The Unit-2 brief provides the following scale and recommends collaborative Planning Poker.

Story Points	Complexity
1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

Illustrative estimates for planning: login – 3 points; add student – 3 points; search student – 2 points; record attendance – 5 points; generate attendance report – 5 points; academic analytics dashboard – 8 points. These are suggested team estimates, not values prescribed by the source document.

9. Sprint Planning and Sprint Backlog

Sprint Planning selects high-priority Product Backlog items that the team believes it can complete in a Sprint. Each Sprint should have a clear goal, selected user stories, tasks, and expected deliverables.

Sprint	Goal	Activities / expected deliverables
Sprint 1	Develop user authentication and project setup.	Initialize repository; build basic user interface; implement registration and login; test authentication.
Sprint 2	Develop Student Management Module.	Implement add student, search student, update student, and delete student features.
Sprint 3	Develop Course and Attendance Management.	Implement add course, assign courses, record attendance, and attendance reports.

After these three example Sprints, the team can plan further Sprints for marks and academic performance, notifications, reporting enhancements, and analytics. Sprint duration should be selected by the team and guide; it is not specified in the supplied brief.

10. Scrum Board Development

A Scrum Board makes work visible and helps the team monitor progress. The Unit-2 brief gives five workflow columns: Backlog, To Do, In Progress, Testing, and Completed.

Backlog	To Do	In Progress	Testing	Completed
Course	Add Student	Search Student	Reports	Project setup
Notifications	Attendance	Student update	Login checks	—

The board should be updated whenever work changes status. A task should move to Completed only after it meets its acceptance criteria and the team's Definition of Done.

11. Scrum Ceremonies

11.1 Sprint Planning

The team selects user stories, estimates effort, defines the Sprint Goal, and creates the Sprint Backlog.

11.2 Daily Scrum

Each member briefly answers: What did I complete yesterday? What will I do today? What challenges or blockers am I facing? The meeting helps coordinate work and identify impediments.

11.3 Sprint Review

At the end of a Sprint, the team demonstrates completed features to stakeholders or the project guide, collects feedback, and considers changes to the Product Backlog.

11.4 Sprint Retrospective

The team discusses what went well, what challenges occurred, and what improvements should be implemented in the next Sprint. The focus is on improving the team's process.

12. Agile Framework Comparison

The Unit-2 brief asks students to compare Scrum, Kanban, Extreme Programming (XP), Lean Development, Large Scale Scrum (LeSS), Scaled Agile Framework (SAFe), and Scrum@Scale. The following concise comparison provides general framework characteristics to support that activity.

Framework	Main approach	Suitability for this mini-project
Scrum	Time-boxed Sprints, defined accountabilities, Product Backlog, and regular review/retrospective.	Strong fit for a student team working in planned increments.
Kanban	Visualizes workflow and limits work in progress; work flows continuously.	Useful for tracking support tasks and ongoing changes.
Extreme Programming (XP)	Emphasizes engineering practices such as frequent testing, pair programming, and continuous integration.	Useful practices can improve code quality, even if the whole framework is not adopted.
Lean Development	Focuses on reducing waste, fast feedback, and delivering value.	Useful as guiding principles for avoiding unnecessary features.
LeSS	Scales Scrum across multiple teams working on one product.	Usually unnecessary for a small student team.
SAFe	Provides a structured approach for coordinating Agile work across large organizations.	Too heavyweight for this mini-project, but relevant for enterprise-scale work.
Scrum@Scale	Connects multiple Scrum teams through coordination structures.	More suitable when several Scrum teams must coordinate.

Recommendation: Scrum is a suitable primary framework for this mini-project because the work can be divided into clear Sprints, responsibilities can be assigned, and working features can be reviewed regularly. The team may also use a simple Kanban-style board to visualize daily task status.

13. Project Timeline and Release Roadmap

The following roadmap aligns the three Sprints described in the Unit-2 brief and extends them logically to cover the remaining listed epics. The exact calendar dates and Sprint duration should be filled in by the project team.

Phase	Planned work	Expected outcome
Sprint 1	Project setup, basic UI, registration, login, authentication testing.	Basic application setup and secure user access.
Sprint 2	Add, search, update, and delete student records.	Working Student Management Module.
Sprint 3	Add courses, assign courses, record attendance, attendance reports.	Working Course and Attendance modules.
Next release	Marks entry, grade calculation, result sheets, notifications, and expanded reports.	Broader academic-management functionality.
Final refinement	Integration testing, usability checks, security review, bug fixing, documentation, and demonstration.	Reviewed and documented mini-project release.

A Burndown Chart can be maintained for each Sprint by recording the remaining work at regular intervals. The chart helps the team see whether the Sprint work is progressing as expected.

14. Testing and Definition of Done

Testing is included in the development activities and Sprint work. Each feature should be checked against its acceptance criteria. A practical Definition of Done for this mini-project is:

- The feature meets its agreed acceptance criteria.
- Input validation and expected error messages are checked.
- The feature is integrated with the relevant application modules.
- Basic functional tests pass and identified defects are addressed or documented.
- The team updates relevant documentation and demonstrates the completed feature.

Examples of checks include valid and invalid login attempts, adding and searching student records, assigning a course, saving attendance, and generating an attendance report. The supplied Unit-2 brief identifies testing as a development-team responsibility but does not prescribe a specific testing tool or test automation framework.

15. Expected Outcomes

- A documented Agile product vision and project scope.
- A Scrum Team structure with role responsibilities.
- Seven Epic descriptions and a collection of user stories.
- Acceptance criteria for important user stories.
- A prioritized Product Backlog and illustrative story point estimates.
- A three-Sprint plan with Sprint goals and deliverables.
- A Scrum Board and a process for Scrum ceremonies.
- A comparison of Agile frameworks and a proposed roadmap.

The expected outcome is a clearly planned Student Record and Academic Management Platform whose features can be developed and reviewed incrementally.

16. Conclusion

This Unit-2 Agile Mini-Project report converts the Student Record and Academic Management Platform concept into Agile planning artifacts. It identifies the product vision, stakeholders, Scrum roles, epics, user stories, acceptance criteria, Product Backlog, story point scale, Sprint plan, Scrum Board, ceremonies, framework comparison, and roadmap.

Using Scrum as the primary approach can help the student team work in manageable increments, review completed features, respond to feedback, and improve its process throughout development.

17. References

1. UNIT-2 AGILE MINI-PROJECT – Student Activities, supplied course/project brief (11 pages).

2. Agile and Scrum terminology used in the report is explained at a general educational level to organize the activities requested in the supplied Unit-2 brief.